

A LOCAL OBSTRUCTION PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

ABSTRACT. We present a local-geometric obstruction proof of the Riemann hypothesis. The two-by-two jet-normalized blocks of the phase kernel of ζ at one or two centers form a finite-dimensional comparison object: a hypothetical off-line zero forces a transverse mode that the actual zeta phase, after a single fixed value-gauge quotient, suppresses at a strictly larger polynomial exponent in the local scale. Two-point and four-point comparisons of this object aggregate over heights via N -point odd-moment cancellation to yield an unconditional zero-free strip around the critical line, hence the Riemann hypothesis. A self-contained final section provides an optional source-energy supply available to the two-point and four-point comparisons if their standalone forms prove insufficient.

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1. INTRODUCTION

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The Riemann hypothesis, originating in Riemann’s 1859 memoir [4], asserts that every non-trivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$; see [1, 2, 3, 5] for surveys. The proof presented in this paper proceeds by a local-geometric obstruction. For each height γ we work on a smooth window I_T centered at $T = \gamma$ of width comparable to $1/Q(\gamma)$, where Q is a scale parameter slowly growing with $|\gamma|$. On this window we fix a real local phase Φ for ζ and study the phase kernel

$$K_\Phi(x, y) = \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, \quad x \neq y,$$

extended by removable singularity at $x = y$. The 2×2 blocks of K_Φ at one or two centers, transformed into the Gram-normalized basis by a fixed elementary matrix P_h , become explicit matrices $J(T)$ and $N_{12}(T_1, T_2)$ whose entries are finite combinations of $q = \Phi'$, q' , q'' , and $\Delta = \Phi(T_1) - \Phi(T_2)$. All subsequent estimates reduce to finite linear-algebraic manipulations of these matrices.

A hypothetical off-line zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ forces a specific transverse mode in the Gram-normalized jet space — the toy mode — which survives the natural value/gauge quotient. The actual zeta phase, under the same quotient, cannot realize that mode: the transverse channel of the actual whitened block is suppressed at a strictly larger polynomial exponent in Q than the toy lower bound. The clash is two-point at minimum and four-point in the full symmetry quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, propagated across heights by an N -point odd-moment cancellation. Aggregation over the dyadic height ladder gives an unconditional zero-free strip around the critical line; combined with standard low-height verification of RH, this yields the Riemann hypothesis.

The argument is self-contained except for one optional input. If either the two-point or the four-point standalone closure proves insufficient, the source-energy lower bound on the canonical source curvature $q''_{\text{can}} = \partial_t^3 Z$, stated as Hypothesis 12.3 in Section 12, supplies the additional polynomial-in- Q room required.

1.1. Active dependency graph. The active spine of the argument is the chain

$$\begin{aligned} &\S 2 \longrightarrow \S 3 \longrightarrow \S 4 \\ &\longrightarrow \S 5, \S 6, \S 7 \\ &\longrightarrow \S 8 \\ &\longrightarrow \S 9 \longrightarrow \S 10 \longrightarrow \S 11. \end{aligned}$$

Section 12 stands aside: it provides the optional source-energy supply consumed by Section 8 when their standalone forms are insufficient. All cross-references to Section 12 from earlier sections are forward references; that section is self-contained.

1.2. Master conditional theorem.

Theorem 1.1 (Master conditional). *Assume the same-point and cross-block jet-limit identities (Lemmas 3.1 and 3.2); the finite-scale whitening loss bound (Lemma 4.3); the toy anomaly visibility lower bound (Theorem 5.3); the two-point or four-point actual-zeta transverse suppression upper bound (Hypothesis 7.1 or Hypothesis 7.3); the corresponding local rigidity theorem (Theorem 8.1 or Theorem 8.4); the N -point odd-moment cancellation (Lemma 9.1); and the aggregation step (Theorem 10.1). Then every nontrivial zero of ζ satisfies $\operatorname{Re} s = \frac{1}{2}$.*

Gap 1.2. The conditional statement is firm; its inputs are the section-level results listed above. The proof is short — a composition of the section-level results — and is written out in Section 11.

1.3. Notation and conventions. We use the symmetric pair convention $T \pm h$ (rather than $T \pm h/2$) throughout. The point-to-jet transform P_h of (2.7) is fixed by this choice. The scale parameter $Q = Q(\gamma)$ is a fixed slowly growing function of the height; all polynomial losses are stated as powers of Q with absolute exponents. For each height parameter T , I_T denotes an interval centered at T of length $|I_T| \ll Q(T)^{-1}$ and contained in $[T/2, 2T]$; high-height analysis takes T sufficiently large that the second inclusion is automatic. “Retained packets” refers to a polynomial-weight sub-family of the local packets at height γ on which the chart-regularity and Gram nondegeneracy hypotheses of Definition 2.3 hold.

2. LOCAL KERNEL AND JET NORMALIZATION

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This section establishes the local geometric machinery used throughout: the Riemann–Siegel phase chart $\Phi_T = \theta$ on the height window I_T of §1.3 (Definition 2.3), the phase kernel K_Φ (Definition 2.6), and the Gram-normalized point-to-jet transform P_h of (2.7).

2.1. The Riemann–Siegel phase chart.

Definition 2.1 (Riemann–Siegel phase). Let $L(z)$ be the unique holomorphic branch of $\log \Gamma z$ on $\Re z > 0$ for which $L(x) = \log \Gamma(x) \in \mathbb{R}$ when $x > 0$. For $t > 0$, define

$$(2.1) \quad \theta(t) := \Im L\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi,$$

and $Z(t) := e^{i\theta(t)} \zeta(1/2 + it)$. The functional equation gives

$$\zeta\left(\frac{1}{2} + it\right) = e^{-i\theta(t)} Z(t), \quad Z(t) \in \mathbb{R} \text{ for } t > 0.$$

Lemma 2.2 (Differentiated theta asymptotics). *Uniformly for $t \in I_T$,*

$$(2.2) \quad q(t) := \theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{48t^2} + O(t^{-4}),$$

$$(2.3) \quad q'(t) = \theta''(t) = \frac{1}{2t} + O(t^{-3}),$$

$$(2.4) \quad q''(t) = \theta'''(t) = -\frac{1}{2t^2} + O(t^{-4}).$$

The constants in the O -terms are absolute.

Proof. Let $z = 1/4 + it/2$. The half-plane $\Re z \geq 1/4$ lies in a fixed Stirling sector. By the differentiated form of Stirling’s expansion, after truncating $\log \Gamma z$ at z^{-3} the remainder and its t -derivatives up to order three are uniform in this sector, with the corresponding differentiated powers of $|z|$. Taking imaginary parts in (2.1) gives the standard Riemann–Siegel expansion

$$\theta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + O(t^{-3}),$$

with a remainder admitting three differentiations in t (needed for $q'' = \theta'''$). Termwise differentiation gives $q(t) = \theta'(t)$; differentiating once and twice more gives $q'(t)$ and $q''(t)$. Since $I_T \subset [T/2, 2T]$, these estimates are uniform on I_T , with absolute constants. $\square$

Definition 2.3 (Local phase chart). A retained high-height local phase chart is the interval I_T together with the real smooth phase

$$\Phi_T(t) := \theta(t) \quad (t \in I_T).$$

When T is fixed by context, $\Phi := \Phi_T$ and $q := \Phi'$. This is a specialized theta chart, not a general phase-chart axiom. The chart conditions used downstream are:

(P1) $\Phi_T = \theta$ on I_T ;

(P2) $q(t) \geq 1$ on I_T , after increasing the global lower-height cutoff if necessary.

If $Q(T) \geq 1$, then (P2) gives the polynomial-in- Q lower bound $q(t) \geq Q(T)^{-c_q}$ with $c_q = 0$. The uniform estimates $q(t) \asymp \log T$, $q'(t) = O(T^{-1})$, and $q''(t) = O(T^{-2})$ on I_T are supplied by Lemma 2.2.

Lemma 2.4 (Phase-derivative lower bound). *Condition (P2) of Definition 2.3 holds for all sufficiently large T . More quantitatively, uniformly on I_T ,*

$$q(t) \geq \frac{1}{2} \log \frac{T}{4\pi} - C T^{-2}$$

for an absolute constant C .

Proof. By Lemma 2.2, $q(t) = \frac{1}{2} \log(t/(2\pi)) + O(t^{-2})$. Since $t \in I_T \subset [T/2, 2T]$, $\log(t/(2\pi)) \geq \log(T/(4\pi))$, and the stated bound follows. The right-hand side is eventually larger than 1. $\square$

Remark 2.5 (No ζ -side regularity is used here). The chart uses ζ only through the identity $\zeta(1/2 + it) = e^{-i\theta(t)} Z(t)$ and through the theta phase. No lower bound on $|Z|$, no zero-free condition in I_T , and no source-energy input is used in Sections 2–3. Such conditions are imposed only in the later sections where actual-zeta transverse suppression or source-energy transfer is consumed.

2.2. The phase kernel.

Definition 2.6 (Phase kernel). For a real C^1 phase Φ on an interval I ,

$$(2.5) \quad K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{q(x)}{\pi}, & x = y, \end{cases} \quad q = \Phi'.$$

Lemma 2.7 (Basic kernel properties). K_Φ is real-valued and symmetric. If $\Phi \in C^1$, the extension across the diagonal is continuous and

$$K_\Phi(T, T) = \frac{q(T)}{\pi}.$$

If $\Phi \in C^4$, the diagonal first and mixed second derivatives exist and are given by Lemma 2.9.

Proof. For $x \neq y$, symmetry follows from $\sin(\Phi(y) - \Phi(x))/(y - x) = \sin(\Phi(x) - \Phi(y))/(x - y)$. Real-valuedness is immediate. For joint continuity across the diagonal, near any $T \in I$ write

$$\Phi(x) - \Phi(y) = (x - y)A(x, y), \quad A(x, y) = \int_0^1 q(y + \tau(x - y)) d\tau.$$

The function A is continuous and $A(T, T) = q(T)$. For $x \neq y$,

$$K_\Phi(x, y) = \frac{A(x, y)}{\pi} \frac{\sin((x - y)A(x, y))}{(x - y)A(x, y)},$$

with the last factor interpreted by its continuous value 1 at zero. Hence $K_\Phi(x, y) \rightarrow q(T)/\pi = K_\Phi(T, T)$ whenever $(x, y) \rightarrow (T, T)$. $\square$

Lemma 2.8 (Off-diagonal kernel derivatives). For $T_1 \neq T_2$ with $s = T_1 - T_2$, $\Delta = \Phi(T_1) - \Phi(T_2)$, $q_j = q(T_j)$, and $\Phi \in C^1$ near T_1, T_2 ,

$$\begin{aligned} K_\Phi(T_1, T_2) &= \frac{\sin \Delta}{\pi s}, & K_x(T_1, T_2) &= \frac{q_1 s \cos \Delta - \sin \Delta}{\pi s^2}, \\ K_y(T_1, T_2) &= \frac{\sin \Delta - q_2 s \cos \Delta}{\pi s^2}, & K_{xy}(T_1, T_2) &= \frac{(q_1 + q_2) s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{\pi s^3}. \end{aligned}$$

Proof. Differentiate $K_\Phi(x, y) = \sin(\Phi(x) - \Phi(y))/\pi(x - y)$ by the quotient and chain rules, then evaluate at (T_1, T_2) . $\square$

Lemma 2.9 (Diagonal kernel derivatives). *If $\Phi \in C^4$ near T , then*

$$(2.6) \quad K_\Phi(T, T) = \frac{q(T)}{\pi}, \quad K_x(T, T) = K_y(T, T) = \frac{q'(T)}{2\pi},$$

$$K_{xy}(T, T) = \frac{q''(T) + 2q(T)^3}{6\pi}.$$

Proof. Set $x = T + u$, $y = T + v$, and write $r = |u| + |v|$; derivatives below are evaluated at T . Factor the phase difference as

$$\Phi(T + u) - \Phi(T + v) = (u - v)A(u, v), \quad A(u, v) = \int_0^1 q(T + v + \tau(u - v)) d\tau,$$

where the integral form extends continuously to $u = v$. Taylor expansion of the integrand gives

$$A(u, v) = q + \frac{q'}{2}(u + v) + \frac{q''}{6}(u^2 + uv + v^2) + O(r^3).$$

For $u \neq v$, and then by continuous extension to $u = v$, $\Phi(T + u) - \Phi(T + v) = (u - v)A(u, v) = O(r)$ gives

$$\frac{\sin(\Phi(T + u) - \Phi(T + v))}{u - v} = A(u, v) - \frac{(u - v)^2}{6} A(u, v)^3 + O(r^3).$$

Keeping terms through total degree two gives

$$K_\Phi(T + u, T + v) = \frac{1}{\pi} \left[q + \frac{q'}{2}(u + v) + \left(\frac{q''}{6} - \frac{q^3}{6} \right) (u^2 + v^2) + \left(\frac{q''}{6} + \frac{q^3}{3} \right) uv + O(r^3) \right].$$

The coefficients of u , v , and uv are respectively $K_x(T, T)$, $K_y(T, T)$, and $K_{xy}(T, T)$, proving the claim. $\square$

2.3. Point-to-jet transform. For symmetric pairs $T \pm h$, define the Gram-normalized coalescing even/odd transform

$$(2.7) \quad P_h = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1/(2h) & 1/(2h) \end{pmatrix} = \text{diag}(\sqrt{2}, 1/\sqrt{2}) \begin{pmatrix} 1/2 & 1/2 \\ -1/(2h) & 1/(2h) \end{pmatrix}.$$

Thus P_h is the literal central value/derivative jet transform followed by the fixed diagonal rescaling $\text{diag}(\sqrt{2}, 1/\sqrt{2})$; at the level of Gram blocks this induces the corresponding diagonal congruence. The value/derivative basis produced by P_h is the *Gram-normalized basis*.

With this convention,

$$(2.8) \quad J(T) = \begin{pmatrix} 2K_\Phi(T, T) & \partial_y K_\Phi(T, T) \\ \partial_x K_\Phi(T, T) & \frac{1}{2} \partial_{xy} K_\Phi(T, T) \end{pmatrix},$$

which is the same-point block computed explicitly in Lemma 3.1.

Remark 2.10 (Pair-separation convention). The pair is $T - h, T + h$, with separation $2h$; the lower row of P_h carries $1/(2h)$. Any formula imported from the alternative $T \pm h/2$ convention or from the literal-jet matrix with lower row $(-1/h, 1/h)$ must be re-conjugated before use.

Remark 2.11 (Convention check). With the literal central value/derivative transform at the pair $T \pm h$, the same-point limit would be

$$\frac{1}{\pi} \begin{pmatrix} q(T) & \frac{1}{2} q'(T) \\ \frac{1}{2} q'(T) & \frac{1}{6} (q''(T) + 2q(T)^3) \end{pmatrix},$$

related to $J(T)$ by the diagonal congruence $\text{diag}(\sqrt{2}, 1/\sqrt{2})$ of (2.7); the $(1, 1)$ entry doubles and the $(2, 2)$ entry is halved. All same-point blocks, cross-blocks, finite-scale blocks, whitening matrices, and quotient covectors in this paper are read in the Gram-normalized convention.

3. JET-LIMIT LOCAL BLOCKS

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From the phase kernel K_Φ and the Gram-normalized transform P_h of Section 2, the small-separation limits of K_Φ at one and two centers produce the same-point Gram block $J(T)$ (Lemma 3.1) and the cross-block $N_{12}(T_1, T_2)$ (Lemma 3.2); for the theta phase, $J(T) \succ 0$ at sufficiently large height (Lemma 3.6).

3.1. Same-point jet limit.

Lemma 3.1 (Same-point jet limit). *Assume $\Phi \in C^5$ near T . Let*

$$A_h(T) = \begin{pmatrix} K_\Phi(T-h, T-h) & K_\Phi(T-h, T+h) \\ K_\Phi(T+h, T-h) & K_\Phi(T+h, T+h) \end{pmatrix}$$

with row and column order $(T-h, T+h)$. Then

$$P_h A_h(T) P_h^\top = J(T) + O(h^2) \quad (h \rightarrow 0),$$

where the $O(h^2)$ is entrywise, with a constant depending only on a fixed neighborhood of T and on a local C^5 -bound for Φ . The limiting block is

$$(3.1) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

Proof. Derivatives below are evaluated at T . Taylor expansion gives

$$\Phi(T \pm h) = \Phi(T) \pm hq + \frac{h^2}{2}q' \pm \frac{h^3}{6}q'' + \frac{h^4}{24}q''' + O(h^5),$$

so

$$\Phi(T-h) - \Phi(T+h) = -2hq - \frac{h^3}{3}q'' + O(h^5).$$

Using $\sin z = z - z^3/6 + O(z^5)$,

$$(3.2) \quad K_\Phi(T-h, T+h) = \frac{q}{\pi} + \frac{h^2}{6\pi}(q'' - 4q^3) + O(h^4).$$

Also $K_\Phi(T \pm h, T \pm h) = q(T \pm h)/\pi$, with

$$q(T \pm h) = q \pm hq' + \frac{h^2}{2}q'' \pm \frac{h^3}{6}q''' + O(h^4).$$

The $O(h^5)$ phase remainder and the $O(h^4)$ q -remainder are controlled by a local C^5 -bound for Φ ; after division by the separation $2h$, the off-diagonal kernel expansion has an $O(h^4)$ remainder, the order needed below since the $(2, 2)$ jet entry divides a second difference by h^2 . For symmetric M , conjugation by (2.7) gives

$$(P_h M P_h^\top)_{11} = \frac{1}{2}(M_{11} + 2M_{12} + M_{22}),$$

$$(P_h M P_h^\top)_{12} = \frac{1}{4h}(M_{22} - M_{11}),$$

$$(P_h M P_h^\top)_{22} = \frac{1}{8h^2}(M_{11} - 2M_{12} + M_{22}).$$

Substituting (3.2) and the Taylor series of $q(T \pm h)$: the constant and odd terms cancel in the second difference, while the first difference cancels its constant and even terms. The divisions by h and h^2 therefore leave entrywise $O(h^2)$ remainders:

$$(P_h A_h P_h^\top)_{11} = \frac{2q}{\pi} + O(h^2),$$

$$(P_h A_h P_h^\top)_{12} = \frac{q'}{2\pi} + O(h^2),$$

$$(P_h A_h P_h^\top)_{22} = \frac{q'' + 2q^3}{12\pi} + O(h^2),$$

which is (3.1). □

3.2. Cross-block jet limit.

Lemma 3.2 (Cross-block jet limit). *Assume $\Phi \in C^5$ in neighborhoods of T_1 and T_2 , let $T_1 \neq T_2$, set*

$$s = T_1 - T_2, \quad \Delta = \Phi(T_1) - \Phi(T_2), \quad q_j = q(T_j),$$

and assume $0 < h < |s|/3$, with h also sufficiently small that the four sampled points lie in the chosen neighborhoods of T_1 and T_2 . Define

$$C_h(T_1, T_2) = \begin{pmatrix} K_\Phi(T_1 - h, T_2 - h) & K_\Phi(T_1 - h, T_2 + h) \\ K_\Phi(T_1 + h, T_2 - h) & K_\Phi(T_1 + h, T_2 + h) \end{pmatrix}$$

with row order $(T_1 - h, T_1 + h)$ and column order $(T_2 - h, T_2 + h)$. Then

$$P_h C_h(T_1, T_2) P_h^\top = \frac{1}{\pi} N_{12}(T_1, T_2) + O(h^2) \quad (h \rightarrow 0),$$

where the $O(h^2)$ is entrywise for fixed $T_1 \neq T_2$; the implied constant may depend on a finite power of $|s|^{-1}$ and on local C^5 -bounds for Φ , but not on h . Here

$$(3.3) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2) s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Proof. The kernel is smooth in a neighborhood of (T_1, T_2) , and the hypothesis $h < |s|/3$ gives $|s + u_1 - u_2| \geq |s|/3$ for $|u_1|, |u_2| \leq h$, bounding the Taylor remainder on the sampled square. Write $K_{ab} = \partial_x^a \partial_y^b K_\Phi(T_1, T_2)$. Taylor expansion to total order four gives, uniformly for $u_1, u_2 \in \{-h, +h\}$,

$$K_\Phi(T_1 + u_1, T_2 + u_2) = \sum_{a+b \leq 4} \frac{K_{ab}}{a!b!} u_1^a u_2^b + O_s(h^5).$$

The subscript indicates that the constant may depend on a finite power of $|s|^{-1}$ and on local C^5 -bounds for Φ in the chosen off-diagonal neighborhoods. For a general 2×2 matrix M ,

$$\begin{aligned} (P_h M P_h^\top)_{11} &= \frac{1}{2}(M_{11} + M_{12} + M_{21} + M_{22}), \\ (P_h M P_h^\top)_{12} &= \frac{1}{4h}(-M_{11} + M_{12} - M_{21} + M_{22}), \\ (P_h M P_h^\top)_{21} &= \frac{1}{4h}(-M_{11} - M_{12} + M_{21} + M_{22}), \\ (P_h M P_h^\top)_{22} &= \frac{1}{8h^2}(M_{11} - M_{12} - M_{21} + M_{22}). \end{aligned}$$

Equivalently, these four expressions apply the sign weights $1, \sigma_2, \sigma_1, \sigma_1\sigma_2$ respectively to the four samples with $u_i = \sigma_i h$. Parity selects the required leading terms: the next surviving polynomial monomials in the $(1, 1)$ entry have total degree at least two, contributing $O(h^2)$; the $(1, 2)$ and $(2, 1)$ numerators select monomials with odd degree in one variable, so after division by h the surviving polynomial corrections are $O(h^2)$; the $(2, 2)$ numerator selects monomials odd in both variables, and division by h^2 again leaves $O(h^2)$. The Taylor remainder $O_s(h^5)$ per sample contributes $O_s(h^5)$ to the $(1, 1)$ entry (no division), $O_s(h^4)$ to the off-diagonal entries (division by $4h$), and $O_s(h^3)$ to the $(2, 2)$ entry (division by $8h^2$). Hence

$$\begin{aligned} (P_h C_h P_h^\top)_{11} &= 2K_{00} + O(h^2), & (P_h C_h P_h^\top)_{12} &= K_{01} + O(h^2), \\ (P_h C_h P_h^\top)_{21} &= K_{10} + O(h^2), & (P_h C_h P_h^\top)_{22} &= \frac{1}{2}K_{11} + O(h^2). \end{aligned}$$

Substituting the off-diagonal derivative formulas of Lemma 2.8 gives (3.3). $\square$

Remark 3.3 (Fixed-separation scope). Lemma 3.2 is an off-diagonal jet limit at fixed separation. It is not a collision expansion for $T_1 \rightarrow T_2$; the constants are not asserted to be uniform as $s = T_1 - T_2 \rightarrow 0$. Any two-center blow-up or near-coalescence analysis must be derived from a separate scaling argument, rather than by citing this lemma with $|s|$ comparable to h .

3.3. Same-point Gram positivity.

Definition 3.4 (Same-point Gram determinant).

$$(3.4) \quad D_J(T) := 4q(T)^4 + 2q(T)q''(T) - 3q'(T)^2.$$

Lemma 3.5 (Algebraic same-point Gram criterion). *For $J(T)$ of (3.1),*

$$(3.5) \quad \text{Tr } J(T) = \frac{2q(T)^3 + 24q(T) + q''(T)}{12\pi}, \quad \det J(T) = \frac{D_J(T)}{12\pi^2}.$$

Moreover,

$$J(T) \succ 0 \iff q(T) > 0 \text{ and } D_J(T) > 0.$$

Proof. The trace and determinant follow by direct computation from (3.1). Since $J(T)$ is real symmetric, Sylvester's criterion gives positive definiteness exactly when its leading principal minor $2q(T)/\pi$ and its determinant are positive. $\square$

Lemma 3.6 (Same-point Gram positivity for the theta phase). *For $\Phi_T = \theta$, and for all sufficiently large retained heights, $J(T)$ is positive definite. More precisely,*

$$(3.6) \quad \lambda_{\min}(J(T)) = \frac{2q(T)}{\pi} (1 + o(1)) = \frac{1}{\pi} \log \frac{T}{2\pi} (1 + o(1)).$$

After increasing the lower-height cutoff if necessary, $\lambda_{\min}(J(T)) \geq 1$; hence the polynomial-in- Q lower bound $\lambda_{\min}(J(T)) \geq Q(T)^{-C_J}$ holds with $C_J = 0$ whenever $Q(T) \geq 1$. The cutoff is qualitative; no explicit numerical lower height is claimed here.

Proof. By Lemma 2.2,

$$q(T) = \frac{1}{2} \log(T/(2\pi)) + O(T^{-2}), \quad q'(T) = O(T^{-1}), \quad q''(T) = O(T^{-2}).$$

Therefore

$$D_J(T) = 4q(T)^4 + O(q(T)T^{-2}) + O(T^{-2}) = 4q(T)^4 (1 + o(1)) > 0$$

for sufficiently large T , and $q(T) > 0$. By Lemma 3.5, $J(T) \succ 0$.

Let $0 < \lambda_1 \leq \lambda_2$ be the eigenvalues of $J(T)$. Since $\lambda_2 \leq \text{Tr } J(T)$,

$$\lambda_1 = \frac{\det J(T)}{\lambda_2} \geq \frac{\det J(T)}{\text{Tr } J(T)} = \frac{D_J(T)}{\pi(2q(T)^3 + 24q(T) + q''(T))} = \frac{2q(T)}{\pi} (1 + o(1)).$$

The Rayleigh quotient at e_1 gives $\lambda_1 \leq J_{11}(T) = 2q(T)/\pi$, proving (3.6). $\square$

Remark 3.7 (Scope of the positivity lemma). Lemma 3.6 is theta-specific. For an arbitrary real phase, the safe statement is only the algebraic criterion $q > 0$ and $D_J > 0$. This lemma does not prove any lower bound for unrelated anisotropy functionals such as $q'' - 6q + 2q^3$, nor does it supply source-energy input of the kind consumed in Section 12.

4. FINITE-SCALE LOCAL BLOCKS AND WHITENING

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

Around a midpoint m at finite separation $s > 0$, the same-point and cross-block jets of Section 3 extend to two same-point Gram blocks $G_{m,\pm}(s)$ and a mixed block $N_m(s)$; whitening these into the dimensionless $\hat{\Omega}_\zeta(s; m)$ (Definition 4.2) with the polynomial-in- Q whitening-loss bound of Lemma 4.3 produces the input consumed by every comparison theorem from Section 5 onward.

4.1. Finite same-point and mixed blocks. Fix a midpoint m and separation $s > 0$; set $t_{\pm} = m \pm s/2$, $q_{\pm} = q(t_{\pm})$, $\Delta_m(s) = \Phi(t_-) - \Phi(t_+)$.

Definition 4.1 (Finite same-point and mixed blocks).

$$(4.1) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

$$(4.2) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+ s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_- s \cos \Delta_m(s)}{s^2} & \frac{(q_- + q_+)s \cos \Delta_m(s) + (2 - q_- q_+ s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}.$$

4.2. Whitened finite block.

Definition 4.2 (Whitened finite block). Provided $G_{m,\pm}(s)$ are positive definite, set

$$(4.3) \quad \hat{\Omega}_{\zeta}(s; m) := G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2}.$$

Lemma 4.3 (Whitening loss). *On the retained packets there is an absolute exponent C_W such that the whitening map has operator-norm distortion at most Q^{C_W} ; equivalently*

$$\|G_{m,\pm}(s)^{-1/2}\|_{\text{op}} \leq Q^{C_W}$$

uniformly in m and s on the retained set.

Gap 4.4. The diagonal of $G_{m,\pm}$ is bounded below by Lemma 3.6; bound the inverse via the explicit 2×2 formula; uniformity in m, s is the nontrivial part. v1 prose to mine: the corrected finite- s holomorphic whitening section.

5. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The next three sections develop the transverse obstruction. The off-line toy phase Φ_{toy} defined below has an explicit transverse mode under the value/gauge projection Π_{trans} of Section 6; the actual zeta phase, under the same Π_{trans} , suppresses the same mode at a strictly larger polynomial exponent (Section 7). The transverse obstruction is the gap between these two bounds.

Definition 5.1 (Off-line toy phase). For $\beta \in (\frac{1}{2}, 1)$ and a window scale Q , the toy off-line phase $\Phi_{\text{toy}}(t; \beta, Q)$ is the model phase of a split pair at offset $\pm(2\beta - 1)$ on a window of size $1/Q$, in the same coordinate and gauge as Definition 2.3.

Gap 5.2. Concrete formula for Φ_{toy} and identification of its q, q', q'' with elementary functions of $(2\beta - 1)$ and the local coordinate. v1 prose to mine: the toy anomaly and transverse obstruction section.

Theorem 5.3 (Toy anomaly visibility). *For every off-line distance $\beta - \frac{1}{2} \neq 0$ at scale Q , the toy whitened block $\hat{\Omega}_{\text{toy}}(s; m)$ satisfies*

$$\|\Pi_{\text{trans}} \mathcal{A}_{\text{toy}}(\hat{\Omega}_{\text{toy}}(s; m))\| \geq Q^{-A_{\text{toy}}}$$

on a positive polynomial-weight subset of midpoints m and separations s , where A_{toy} is an absolute exponent and $\mathcal{A}_{\text{toy}}$ is the toy anomaly functional defined explicitly from the entries of $\hat{\Omega}_{\text{toy}}$.

Gap 5.4. Definition of $\mathcal{A}_{\text{toy}}$ (a polynomial combination of the entries of $\hat{\Omega}_{\text{toy}}$ selected to vanish at $\beta = \frac{1}{2}$ but not for $\beta \neq \frac{1}{2}$) and the lower bound on retained (m, s) . The bound is to be insensitive to the value channel and the residual gauge.

6. TRANSVERSE PROJECTION

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The transverse projection Π_{trans} is the fixed value/gauge quotient under which the toy whitened block of Section 5 and the actual-zeta whitened block of Section 7 are compared. Its construction is the most sensitive structural decision in the rebuild: Π_{trans} is fixed here and is not to be re-chosen afterward.

Definition 6.1 (Transverse projection). Let $\mathcal{V}_{\text{val}}$ denote the finite-dimensional subspace of the Gram-normalized jet space spanned by the value, gauge, and background directions to be quotiented out. A transverse projection Π_{trans} is an operator onto the orthogonal complement of $\mathcal{V}_{\text{val}}$ satisfying, on the retained packets:

- (II1) Π_{trans} annihilates $\mathcal{V}_{\text{val}}$ exactly;
- (II2) Π_{trans} is bounded with polynomial-in- Q loss;
- (II3) Π_{trans} is invariant under the residual gauge freedom left in Φ after Definition 2.3.

Gap 6.2. The choice of $\mathcal{V}_{\text{val}}$ must precede Theorem 5.3 and Hypotheses 7.1 and 7.3; it is not to be re-chosen after any has been invoked. v1 prose to mine: the leading value matrix and canonical calibration section, together with the local projective rigidity of the one-pair and multi-pair families section.

7. ACTUAL-ZETA TRANSVERSE SUPPRESSION

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The actual zeta phase Φ defines two whitened blocks — the two-center block $\widehat{\Omega}_{\zeta}(s; m)$ and a four-center analogue $\widehat{\Omega}_{\zeta}^{(4)}$ for the symmetric quartet — on which the actual-zeta anomaly is suppressed at a strictly larger polynomial exponent than the corresponding toy lower bound of Theorem 5.3. This is the load-bearing asymmetry of the program: the off-line scenario forces the projective anomaly to be large; the actual zeta phase forces it to be small. The two-point and four-point comparisons of Section 8 consume different forms of this suppression, stated separately below.

7.1. Two-point suppression.

Hypothesis 7.1 (Two-point actual-zeta transverse suppression). For the two-center packet associated with ρ and a symmetry partner, after the gauge-fixed normalization of Definition 2.3 and the projection Π_{trans} of Definition 6.1,

$$\|\Pi_{\text{trans}} \mathcal{A}_2(\widehat{\Omega}_{\zeta}(s; m))\| \leq Q^{-B_{\zeta,2}}$$

for an absolute exponent $B_{\zeta,2} > A_{\text{toy}}$ on the retained subfamily used in Theorem 5.3. Here $\mathcal{A}_2$ is the restriction of the toy anomaly functional of Theorem 5.3 to the actual-zeta two-center whitened block.

Gap 7.2. The two-point case of the structural gap. Consumed by Theorem 8.1. Admissible proof routes: (i) a standalone route from the local jet-Gram geometry alone; (ii) an energy-supply route consuming the source bound of Section 12 via Lemma 12.5. v1 prose to mine: the odd holomorphic transverse channel and corrected finite- s holomorphic whitening sections.

7.2. Four-point suppression.

Hypothesis 7.3 (Four-point actual-zeta transverse suppression). For the four-center packet associated with the symmetry quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, under the four-point projection $\Pi_{\text{trans}}^{(4)}$ extending Definition 6.1,

$$\|\Pi_{\text{trans}}^{(4)} \mathcal{A}_4(\widehat{\Omega}_{\zeta}^{(4)})\| \leq Q^{-B_{\zeta,4}}$$

for an absolute exponent $B_{\zeta,4} > A_{\text{toy}}^{(4)}$ on the retained four-point subfamily. Here $\mathcal{A}_4$ is the four-point analogue of the toy anomaly functional and $A_{\text{toy}}^{(4)}$ is the four-point toy visibility exponent.

Gap 7.4. The four-point case of the structural gap. Consumed by Theorem 8.4. Same admissible proof routes as Hypothesis 7.1. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, four-point variant.

Remark 7.5 (Standalone and energy-supply routes). Hypotheses 7.1 and 7.3 are stated independently of the technology of their proofs. A purely local proof would use only the jet-Gram and quotient geometry of Sections 3–6; an energy-supply route may use the source-energy module of Section 12. Forward references from Section 8 indicate which hypothesis is invoked, not which route proves it.

8. LOCAL PROJECTIVE RIGIDITY

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

Theorem 5.3 provides the toy lower bound and Hypotheses 7.1 and 7.3 provide the matching actual-zeta upper bounds on the transverse anomaly under Π_{trans} of Definition 6.1. The present section assembles the comparison. On the two-point family attached to ρ and a single symmetry partner, and on the four-point family of the full symmetric quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, the exponent inequality $A_{\text{toy}} < B_{\zeta}$ (resp. its four-point analogue $A_{\text{toy}}^{(4)} < B_{\zeta}^{(4)}$) is incompatible with the existence of the configuration. The two cases share the comparison mechanism but differ in the underlying jet-Gram object: the two-point case uses the cross-block N_{12} of Lemma 3.2; the four-point case uses a four-center analogue $\hat{\Omega}_{\zeta}^{(4)}$ and a four-point extension $\Pi_{\text{trans}}^{(4)}$ of the projection of Definition 6.1. Either case suffices for the aggregation step in Section 10.

8.1. Two-point comparison. The two-point comparison takes ρ and one symmetry partner (typically $\bar{\rho}$ or $1 - \rho$) and reads off the joint contribution to $N_m(s)$ at matched midpoint and separation. The exponent inequality $A_{\text{toy}} < B_{\zeta}$ excludes the pair.

Theorem 8.1 (Two-point rigidity). *Let $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ sufficiently large, and let (T_1, T_2) be the height pair attached to ρ and one of its symmetry partners. Under Lemmas 3.1, 3.2, 4.3, Theorem 5.3, and Hypothesis 7.1,*

$$A_{\text{toy}} < B_{\zeta,2} \quad \implies \quad \rho \notin \{\zeta = 0\}.$$

Gap 8.2. Compose Theorem 5.3 (lower bound) with Hypothesis 7.1 (upper bound) at matched (m, s) ; show the exponent gap survives the polynomial whitening loss (Lemma 4.3) and the polynomial Gram lower bound (Lemma 3.6). The proof is short — arithmetic on absolute exponents — once the inputs are in place.

Remark 8.3 (Energy-supply route, if invoked). If Hypothesis 7.1 cannot be discharged from the local jet-Gram geometry alone for the two-point case, the source-energy lower bound Hypothesis 12.3 of Section 12, transferred via Lemma 12.5, supplies the missing polynomial-in- Q room. Theorem 8.1 is then conditional on that source as well as on the local inputs.

8.2. Four-point comparison in the symmetric quartet. The four-point comparison uses the full symmetry packet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$. The four-center jet-Gram block carries strictly more rigidity than the two-point cross-block: the toy visibility lower bound holds in a refined quotient invisible to the two-point comparison, and the matching suppression hypothesis is correspondingly stronger. The four-point theorem is stated independently of Theorem 8.1.

Theorem 8.4 (Four-point rigidity). *Let ρ be a hypothetical off-line zero and $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ its symmetry quartet. Under the four-point analogue of Theorem 5.3 and Hypothesis 7.3,*

$$A_{\text{toy}}^{(4)} < B_{\zeta,4} \quad \implies \quad \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\} \not\subset \{\zeta = 0\}.$$

Gap 8.5. State and prove the four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ of $\widehat{\Omega}_\zeta$; the four-point projection $\Pi_{\text{trans}}^{(4)}$; and the four-point analogue of Theorem 5.3 matching Hypothesis 7.3. Assemble the exponent inequality. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, together with the corrected fixed-target quotient package.

Remark 8.6 (Energy-supply route, if invoked). The four-point analogue of Remark 8.3 applies: if Hypothesis 7.3 cannot be proved from local geometry alone, Hypothesis 12.3 of Section 12 supplies the missing room via the four-point branch of Lemma 12.5, separating $A_{\text{toy}}^{(4)}$ from $B_{\zeta,4}$.

9. N -POINT ODD-MOMENT CANCELLATION

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The two-point and four-point comparisons of Sections 8.1 and 8.2 carry finite Taylor remainders of Φ at packet centers. An explicit discrete N -point measure $\mu_{N,Q}$ on s -space, with its low-order odd moments cancelled by construction, absorbs those remainders below $\min(Q^{-A_{\text{toy}}}, Q^{-B_\zeta})$ uniformly in the height.

Lemma 9.1 (N -point odd-moment cancellation). *Let $\widehat{\mu}_{N,Q}$ be the symmetric N -point packet measure attached to height T and scale Q . For every $r \geq 0$,*

$$(9.1) \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0.$$

The same identity holds with $\widehat{\mu}_{N,Q}$ replaced by its convolution with any even bump $\widehat{\phi}$: for every $r \geq 0$,

$$(9.2) \quad \partial_z^{2r+1} (\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z)) \Big|_{z=0} = 0.$$

Gap 9.2. Pure parity argument; v1 carries the full proof in the N -point odd-moment cancellation section. Migration is straightforward provided the symmetry packet of (9.1) matches Definition 2.3.

10. AGGREGATION AND ZERO-FREE STRIP

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

The two-point and four-point rigidity theorems exclude any single off-line zero above height T_0 ; the upgrade to a zero-free strip proceeds by aggregating along the dyadic height ladder, exploiting the polynomial nature of every loss in Lemmas 4.3, 3.6, Theorem 5.3, and Hypotheses 7.1 and 7.3.

Theorem 10.1 (Zero-free strip). *There is $T_0 > 0$ such that $\zeta(\beta + i\gamma) \neq 0$ for every $\beta \neq \frac{1}{2}$ and $|\gamma| \geq T_0$.*

Gap 10.2. Aggregate Theorem 8.1 or Theorem 8.4 across the dyadic ladder; the polynomial losses combine into a single polynomial-in- $Q(\gamma)$ factor, and the exponent inequality $A_{\text{toy}} < B_\zeta$ carries through uniformly.

11. THE RIEMANN HYPOTHESIS

paper: [LIVE] v1: [n/a] referee: [not-started] sympy: [n/a] sim: [n/a] lean: [not-started]

The dyadic zero-free strip of Theorem 10.1 excludes any off-line zero above the absolute height cutoff T_0 ; at heights below T_0 the standard numerical verification of RH covers the remaining zeros, completing the unconditional statement.

Theorem 11.1 (Riemann hypothesis). *Every nontrivial zero of ζ satisfies $\text{Re } s = \frac{1}{2}$.*

Proof. By Theorem 10.1, no off-line zero can lie at height $|\gamma| \geq T_0$; the standard low-height numerical verification of RH (see, e.g., [3, 5]) covers the range $|\gamma| < T_0$. $\square$

12. SOURCE-ENERGY SUPPLY (OPTIONAL)

paper: [LIVE] v1: [not-started] referee: [not-started] sympy: [not-started] sim: [not-started] lean: [not-started]

This section is self-contained: its sole role is to supply a single source-energy lower bound that Theorems 8.1 and 8.4 may invoke when their standalone forms in Section 8 are insufficient. Nothing earlier in the paper depends on this section.

12.1. Canonical source curvature.

Definition 12.1 (Canonical source curvature). The canonical source curvature of ζ on the chart of Definition 2.3 is

$$(12.1) \quad q''_{\text{can}} := \partial_t^3 Z(t),$$

where $Z(t)$ is the real critical-line scalar of ζ in the same chart.

Gap 12.2. Identification of q''_{can} with $\partial_t^3 Z$ is a convention; chart-regularity and zero-separation hypotheses of Definition 2.3 carry over verbatim. v1 prose to mine: the upstream Gate-1 arithmetic inputs section.

12.2. Source-energy bound.

Hypothesis 12.3 (Source-energy lower bound). On retained packet intervals $I \subset I_T$, the canonical source curvature satisfies

$$(12.2) \quad \int_I |q''_{\text{can}}(m)|^2 dm \geq Q^{-A_E} |I|$$

on a positive polynomial-weight subfamily, for an absolute exponent A_E .

Gap 12.4. polynomial lower bound on $\partial_t^3 Z$ across the retained packet family. This is an analytic estimate on ζ and not on the local jet-Gram geometry; it is what makes Section 12 optional. v1 prose to mine: the Hardy-curvature mass-good packet source block.

12.3. Energy-to-suppression transfer.

Lemma 12.5 (Source energy to transverse suppression). *Hypothesis 12.3, after carrier decomposition and packet trace normalization, implies one of Hypothesis 7.1 (with exponent $B_{\zeta,2} = B_{\zeta,2}(A_E) > A_{\text{toy}}$) or Hypothesis 7.3 (with exponent $B_{\zeta,4} = B_{\zeta,4}(A_E) > A_{\text{toy}}^{(4)}$), according to which carrier transfer is established into the corresponding transverse channel of $\widehat{\Omega}_{\zeta}(s;m)$ or $\widehat{\Omega}_{\zeta}^{(4)}$.*

Gap 12.6. Carrier decomposition and packet trace step taking $\int |q''_{\text{can}}|^2$ on the source side into a lower bound on $\Pi_{\text{trans}} \widehat{\Omega}_{\zeta}$ (two-point case) or $\Pi_{\text{trans}}^{(4)} \widehat{\Omega}_{\zeta}^{(4)}$ (four-point case). v1 prose to mine: the certified downstream status after the canonical source-graded closure section, and the carrier-corrected source bridge package.

APPENDIX A. ARCHIVE MAP

paper: [DIAGNOSTIC] v1: [n/a] referee: [n/a] sympy: [n/a] sim: [n/a] lean: [n/a]

The previous large draft is preserved in place at `rh/proof_of_rh.tex` and is treated as a quarry from which prose and computations may be migrated, after restatement and refereeing, into the present file. Pointers to relevant v1 sections are given inline at each `wip` mark above; the table below summarizes the principal mining correspondences.

Rebuild section	v1 source (in proof_of_rh.tex)
§2	Local kernel and jet normalization.
§3	Finite- s local blocks; jet-limit local blocks.
§4	Corrected finite- s holomorphic whitening.
§5	Toy anomaly and transverse obstruction.
§6	Leading value matrix and canonical calibration.
§7	Odd holomorphic transverse channel; corrected fixed-target quotient package.
§8	Local projective rigidity of the one-pair and multi-pair families; finite coefficient table contracts.
§9	N -point odd-moment cancellation.
§10, §11	Master conditional theorem; introduction.
§12	Hardy-curvature mass-good packet source block; carrier-corrected source bridge package; upstream Gate-1 arithmetic inputs.

The v1 sections “Endgame architecture and remaining propagation problem outside the clean mixed four-point case,” “Downstream local closure package: Gate 2, wall escape, and carrier rows,” “Unified finite-survival architecture,” “Current proof state after the local package audits,” “How the older framing maps to the new framing,” and “No-go and audit ledger” are deliberately not migrated; their content is route history and process bookkeeping rather than active proof material.

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